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Key Points:

- Clear air turbulence in a real case is reproduced by the use of a numerical weather prediction model
- If the resolution is fine enough for a large eddy simulation (LES), turbulent eddies associated with Kelvin-Helmholtz instability are reproduced
- Turbulence eddies are shown to be realistic because virtual flight data in the LES data agree well with real flights

Supporting Information:

Supporting Information may be found in the online version of this article.

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Clear Air Turbulence Resolved by Numerical Weather Prediction Model Validated by Onboard and Virtual Flight Data

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Abstract A clear air turbulence (CAT) occurred on 30 December 2020 over Tokyo, Japan. The CAT was largely generated by breaking Kelvin-Helmholtz (KH) instability waves in the free atmosphere. A regional numerical weather prediction model simulated the event with fine resolution (35 m). Onboard-recorded flight data and a flight simulation were utilized to validate the meteorological simulation. The locations of the reproduced strong turbulence agree well with the regions where flights encountered turbulence on that day. In a simulation with the finest resolution, the KH waves and their breaking were resolved. When the resolution was finer, the turbulent eddies were stronger, causing meteorological effects on the airplanes. The response of a virtual airplane to the simulated turbulence was estimated using a flight simulation. By comparing onboard-recorded data with virtual flight data, we confirm that turbulent eddies are reasonably reproduced.

Plain Language Summary Clear air turbulence (CAT) is a threat to aviation safety. CAT occurs outside clouds, making it difficult to detect by pilots and onboard radars. When an airplane encounters strong CAT, changes in aerodynamic forces and moments cause significant shaking of the airplane. Although CAT generation in the free atmosphere has been studied by high-resolution numerical simulations, few studies simulated aircraft-scale turbulence eddies and validated them with high-frequency airborne observation. We investigate a past CAT event in Japan using a very high resolution (35 m) simulation based on a regional weather prediction model of daily weather forecasts. The simulated turbulence was verified by comparing the virtual flight in simulated wind fields with realistic flight data recorded by airplanes during the event. Simulated aircraft responses in finer weather simulations match flight records better.

1. Introduction

Turbulence in the free atmosphere has many causes, such as jet streams, gravity waves, cumulus convection, fronts (Bramberger, 2019; Bramberger et al., 2018; Kim & Chun, 2010; R. D. Sharman et al., 2012; Trier et al., 2012; Trier et al., 2020). Particular attention should be paid to turbulence in the clear air (e.g., clear air turbulence [CAT]) because it has no visual signs. Turbulence (including CAT) in the free atmosphere is critical for aviation safety (Japan Transport Safety Board, 2014).

Invisible CAT has been visualized using numerical simulations with fine resolution, that is, large eddy simulation (LES), that can resolve grid-scale turbulent eddies. Several studies have investigated turbulence in the free atmosphere by LES in idealized cases (Kudo, 2013; Lane & Sharman, 2014; Zovko-Rajak & Lane, 2014). A realistic case was addressed by Lane et al. (2003) by performing LES with resolution of 27.8 m to investigate turbulence near the top of convective clouds with the computational boundaries forced by the outputs of a regional scale model. Another example of a LES study is Yoshimura et al. (2022), which used a 3D weather simulation with 250 m resolution and demonstrated the impact of simulated turbulence on a large airplane. Owing to the fine resolution of LES, these cases reproduced the turbulence eddies that accompany large wind fluctuations.

For a finer resolution, larger fluctuations appear as previously demonstrated by these LES studies. However, it is still uncertain how well turbulent eddies can be realistically reproduced and whether they are a threat to aircrafts. It is preferable to use observational data to verify LES, although there are few intensive observations for turbulence in the free atmosphere. For comparison, LES must also consider a wide variety of spatio-temporal atmospheric

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variables, which have been treated by regional numerical weather prediction (NWP) models. State-of-the-art supercomputers allow an LES to be performed by using an NWP model with a numerical domain that is large enough to include spatial heterogeneities.

This study addresses a CAT event that affected aircrafts by straightforwardly advancing the resolution of a regional NWP model and the numerical results are verified by the aircrafts themselves. We perform a 35 m resolution LES by using the most powerful Japanese supercomputer, Fugaku, and the resulting wind fields are verified by a flight simulator and by flight data, which include high-frequency observation of winds and aircraft acceleration, on the day in question. During 7:00–11:00 UTC on 30 December 2020, several encounters with CAT were reported near Tokyo, Japan (hereafter, “target event”). Most flights were traveling on the standard landing paths to Tokyo International (Haneda) Airport.

The surface weather map of several hours after the target event is shown in Figure 1a. Surface low pressure existed to the east of Japan. As implied by the peculiar local curve of the isobar (1,020 hPa) over Tokyo, the cold advection appeared to be blocked by the central mountain range of Japan and directional vertical shear (i.e., backing) could be enhanced over Tokyo.

Figure 1b shows where aircrafts encountered turbulence during the target event: the location of each moderate or greater turbulence report in JAL PIREP (Japan Airlines Pilot Report) data is shown. A person may find it difficult/impossible to stand if aircrafts encounter moderate/severe turbulence. A total of 49 encounters (32 with weather reports) were reported in this period, and two flights reported “severe” turbulence (09:57, 10:03 UTC). The majority of weather reports showed clear air. The altitude histogram of the PIREP data (Figure 1c) shows that most encounters were reported at 3 km, which is in the free atmosphere.

The remainder of this paper is organized as follows. The atmospheric and flight simulations and the flight data are explained in Section 2. The simulated CAT and comparisons with the flight records are detailed in Section 3.1. The LES results are verified using flight simulations and data in Section 3.2. Finally, the paper concludes with Section 4.

2. Methods and Data

2.1. Numerical Weather Prediction Model

To simulate CAT in the free atmosphere for the target event, we employ a meso-scale NWP model. We use ASUCA (A System based on a Unified Concept for Atmosphere), developed by the Japan Meteorological Agency (JMA) for operational regional weather predictions (Hara et al., 2012; Ishida et al., 2009, 2010, 2022). The full physical processes are considered. Horizontal resolution, dx , is 2 or 5 km for the operational forecasts, while we use a much finer dx to resolve turbulent eddies.

We conduct numerical simulations for the target event with horizontal resolutions of $dx = 500, 250, 70$, and 35 m. Figure 2 shows the computational domains. We refer to the simulations as D1, D2, D3, and D4, respectively. Initial and boundary conditions for D1 are generated based on JMA-MSM analysis data (given every 3 hr with 5 km resolution, Japan Meteorological Agency (2013)). The D1 simulation is initialized for 03:00 UTC on 30 December 2020. Each child simulation is initialized an hour later than the parent simulation and is dynamically downscaled via one-way nesting.

D1–D3 have 112 levels below the model top at 17.2 km with vertical grid intervals fixed at a constant $dz = 100$ m between 1 and 6.3 km. D4 has 150 levels below the model top and the vertical grid intervals gradually increase from the surface to 160 m, remaining at a constant $dz = 35$ m below 4 km, and increase again until the model top at 15.7 km. For all domains, Deardorff sub-grid scale scheme governed the turbulent mixing. Rayleigh damping is used in top 10 layers, while cumulus parameterization is not used in any simulations. Each time-integration is performed until 12:00 UTC, and all simulations took a total of 4.5 weeks on 384 nodes of the Fugaku supercomputer.

2.2. Flight Simulation

A 2D flight simulation used in our previous paper (Yoshimura et al., 2022) is applied to the present NWP results. The simulation solves the motion of the Boeing 787 (B787) model but does not consider the pitching rate control embedded in the aircraft in reality, because the details are not disclosed. The simulated wind speeds are inputted into the flight simulation, which computes additional aerodynamic forcing. The simulation uses a fourth order Runge–Kutta time integration with $dt = 0.1$ s. Detailed explanation of the flight simulation and the impact not

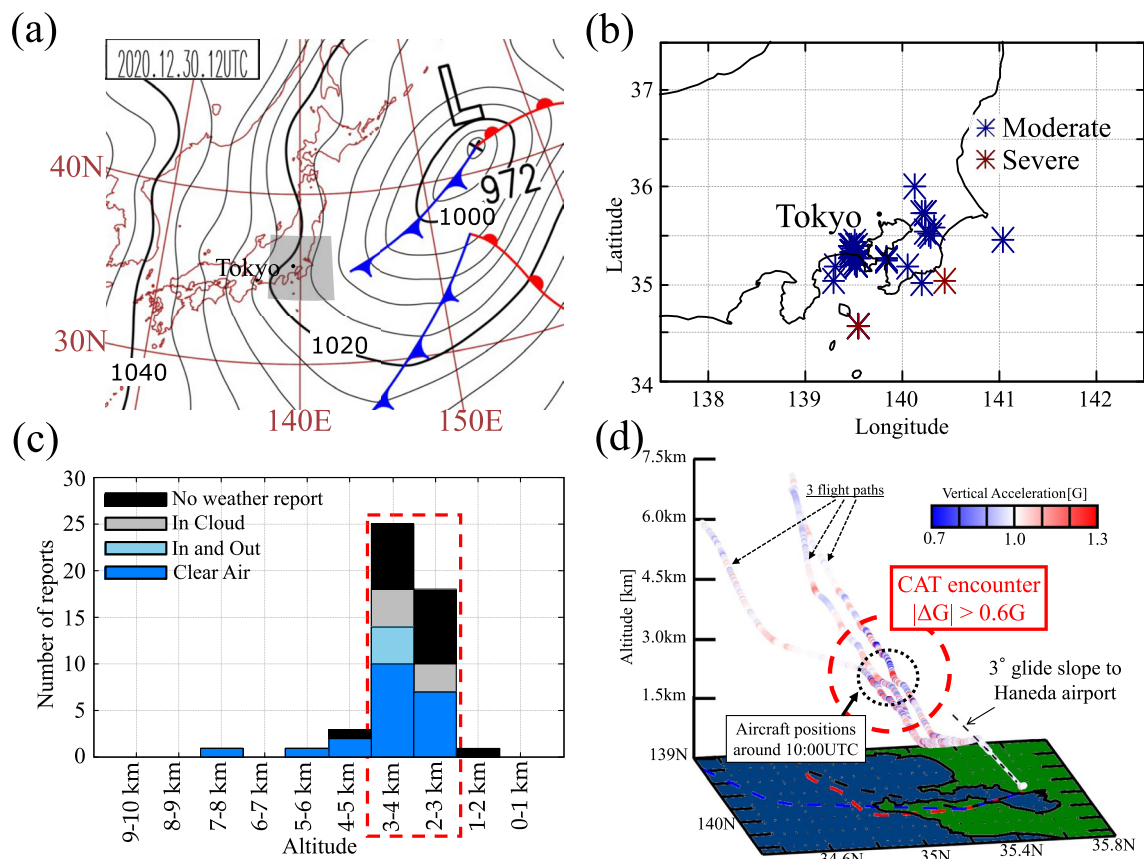


Figure 1. (a) Surface weather map at 12:00 UTC on 30 December 2020 provided by Japan Meteorological Agency with region of (b) visualized as the dark region. (b) The PIREP data during the target event. (c) Altitude histogram of the reports of encounters with turbulence in clouds (In Cloud), in and out of clouds (In and Out) and in clear air. (d) Three trajectories of airplanes flying near the locations of “severe” turbulence in (a).

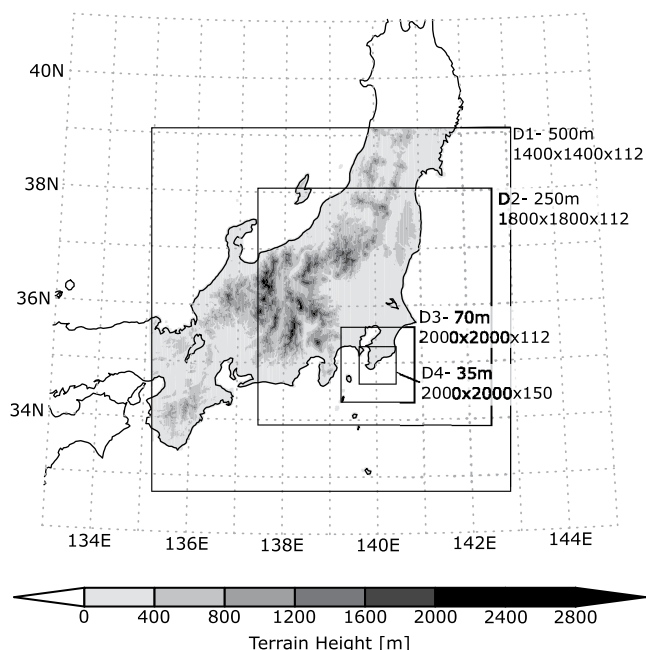


Figure 2. Computational domains for the ASUCA simulations.

considering the pitching rate control has on the results are described in Text S3 in Supporting Information S1.

2.3. Flight Records

We obtain onboard flight records from three airplanes (B787 $\times$ 2, B767 $\times$ 1) that encountered turbulence during the target event. Latitude, longitude, airspeed, and horizontal winds are recorded at 1-s intervals, except for g-histories, which are recorded at 1/16 (B787) or 1/8-s (B767) intervals. We utilize these flight records to validate the simulated results and discuss the situation of the target event. Figure 1d shows the three flight paths with colors representing the vertical accelerations. All flights were bound for Tokyo International Airport and were significantly shaken (a maximum of 0.86 g) at around 3 km. Airplanes were flying at around the ground speed of 130 m/s during the encounter. Eddy dissipation rate estimated from one of the flight data is presented in Text S4 in Supporting Information S1.

3. Results and Discussions

3.1. Results of Atmospheric Simulation

3.1.1. Environment of Clear Air Turbulence (Simulation D1)

Figures 3a and 3b are a horizontal cross-section at 3 km altitude and a vertical cross-section along line XX' shown in (a), respectively, and in simulation

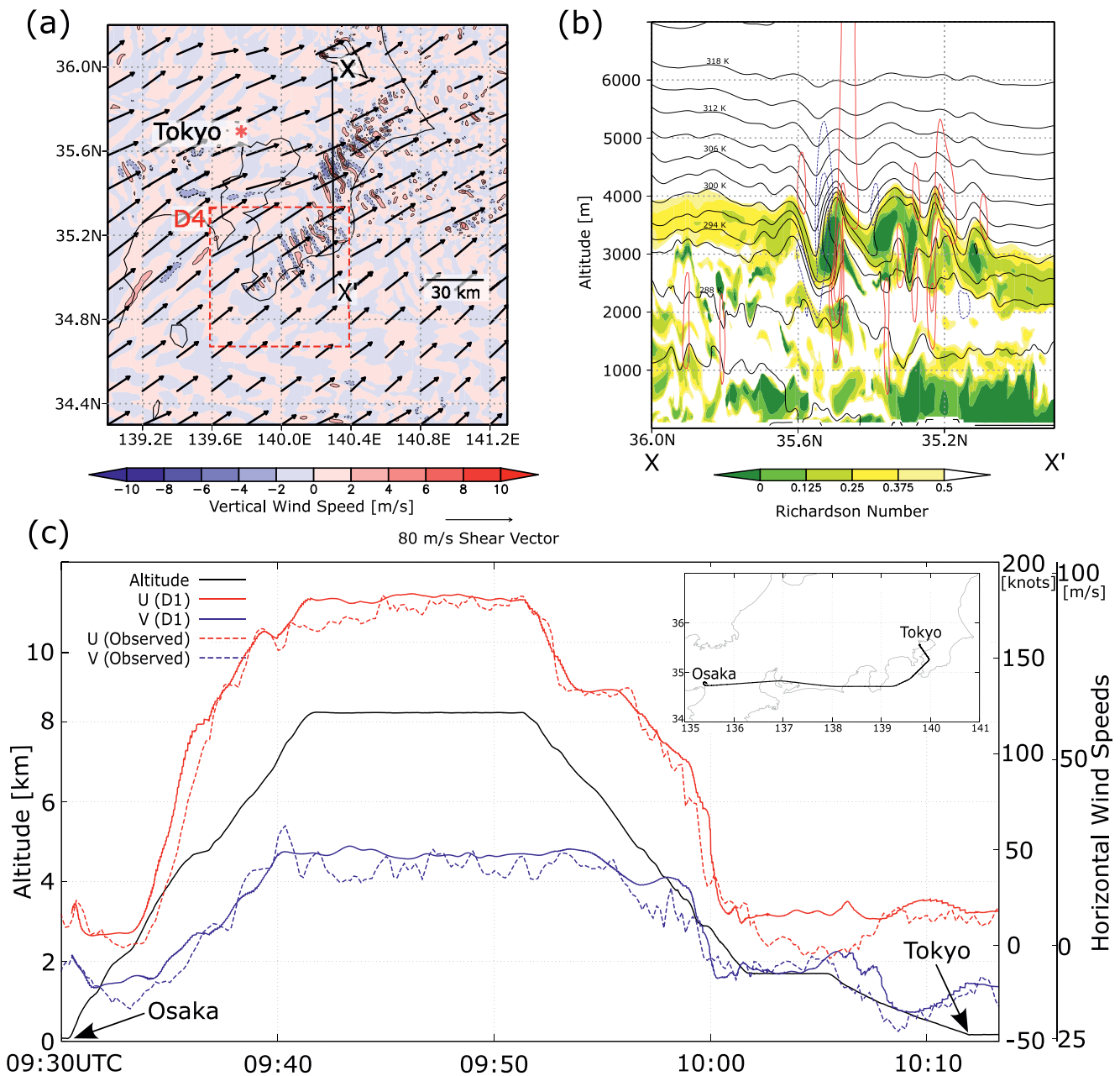


Figure 3. (a) Horizontal cross-section of the D1 simulation at 3 km altitude at 10:00 UTC on 30 December 2020. Waves are found near Tokyo (the red dot). The red rectangle shows the boundary of the D4 domain. (b) Vertical cross-section XX' shows the Richardson number, potential temperature [K], and vertical wind speed at 2 m/s intervals (red: updraft blue: downdraft). (c) Comparison between the flight record and the D1-simulated wind speeds on the flight path. Red/blue curves are the comparison between U (west-east wind)/V (south-north wind).

D1 at 10:00 UTC on December 30, when many MOG-level PIREPs were reported. Wave trains with couplets of updrafts (red) and downdrafts (blue) are significant in the east part of Tokyo (Figure 3a). Isophase lines of the waves are almost perpendicular to the shear vectors (black), which are the differences between wind vectors at 4 km and those at 2.5 km. Waves are caused by the KH instability in the vertical shear. Total cloud condensate is quite small in the region of the shear, which suggests that KH waves occur in the clear air (details provided in Text S1 in Supporting Information S1). The region where the waves occur coincides with the locations of many PIREPs (Figure 1b). In Figure 3b, the area with a Richardson number (Ri) smaller than the critical value of 0.25 (Drazin & Reid, 1981; Nappo, 2002) is found at 3 km, between 35.2 and 35.6 N. Larger amplitudes of updrafts/downdrafts occur in the area.

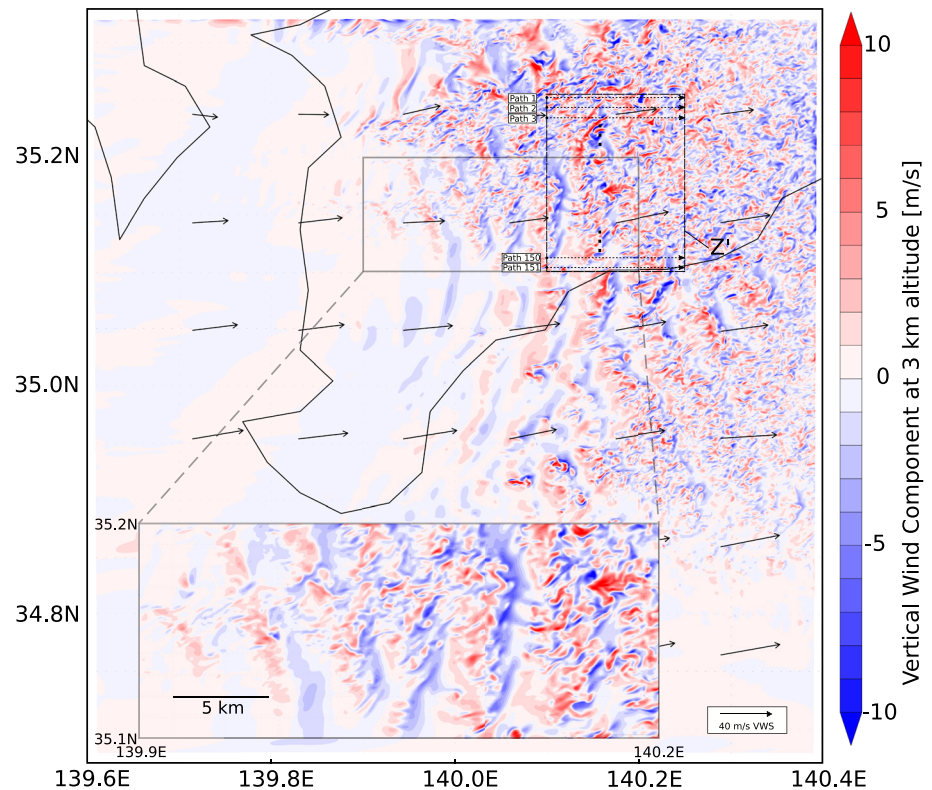


Figure 4. Horizontal cross-section of the D4 simulation at 3 km altitude at 10:00 UTC on 30 December 2020. Shear vectors are shown as black vectors. Rectangle Z' is the region where the virtual flights were performed.

Figure 3c compares horizontal winds in simulation D1 with the wind records from one of the flights shown in Figure 1d. Simulation D1 reasonably reproduces the time series of the recorded winds from 9:30 UTC when the airplane took off from Osaka to 10:12 UTC when it landed in Tokyo. The aircraft descended through the environment with strong vertical shear around 10:00 UTC in the region with small Ri.

3.1.2. Resolved Clear Air Turbulence (Simulation D4)

In simulation D1 with $dx = 500$ m, significant large-scale fluctuations in vertical velocity are found in the region with small Ri. However, the maximum amplitude of $\sim 6 \text{ m s}^{-1}$ with wavelengths much larger than aircraft-size are not expected to affect flight maneuvers. The realistic pictures of CAT can be identified using simulations with finer resolution. This subsection focuses on simulation D4, performed with the finest resolution.

A 3 km horizontal cross-section of simulation D4 at 10:00 UTC on 30 December 2020 is shown in Figure 4. The couplets of updrafts and downdrafts with the spacing of 4–5 km align with a direction similar to Figure 3a, from west-southwest to east-northeast. Since particularly large shear vectors are present as similar to D1, the broad structures of KH waves are also reproduced. The KH waves grow from the west and are organized in 6–8 crests along with 35.05 N in Figure 4 (also seen at 10:11 UTC, Figures 5a and 5b). When the amplitudes exceed 10 m s^{-1} , they break up into smaller-scale turbulent eddies, as displayed in the small panel.

We can infer the scenario when an aircraft encountered the CAT by simulation D4 and the onboard-recorded flight data. Figure 5c shows the flight path and the vertical acceleration of an airplane whose flight records are shown in Figure 3c. The airplane experienced small ($\pm 0.1 \text{ g}$) and low-frequency oscillations at 4–5 km first, followed by intense ($\pm 0.5 \text{ g}$) and high-frequency shaking at 3–4 km. The low-frequency oscillation at 4–5 km is thought to be induced by the growing, organized KH waves (139.9E–140.1E in Figure 5b). The subsequent high-frequency shaking at 3–4 km is caused in the area where the KH waves break (140.15 E–140.25 E in Figure 5b). The small plot in Figure 5c shows the horizontal wind speed and vertical acceleration observed by the airplane between 9:56 and 9:58 UTC. For the low-frequency oscillation found around 9:56:30 UTC, we estimate its spatial scale to be 4.63 km, which agrees well with the wave-length of KH waves in simulation D4 (4–5 km).

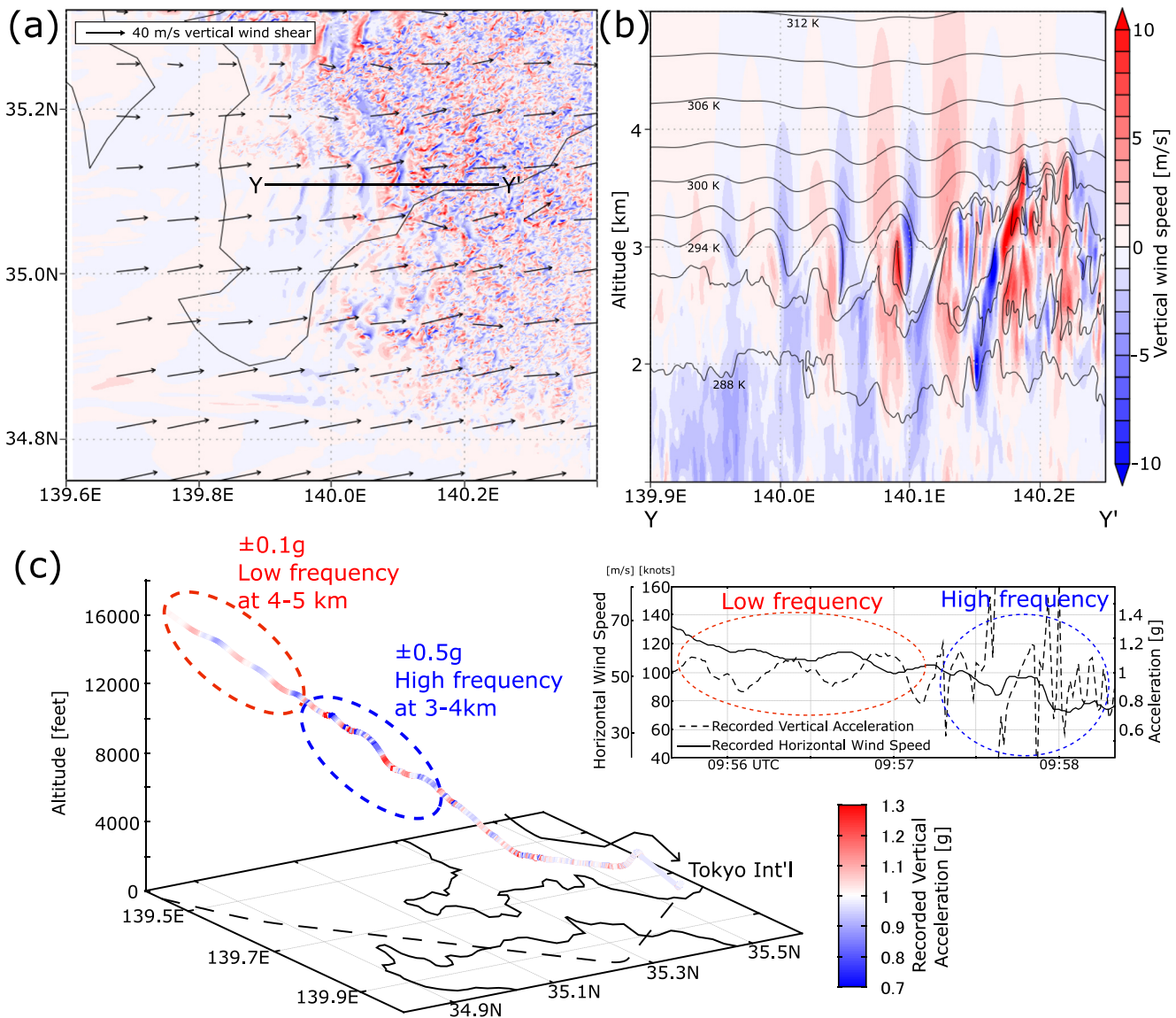


Figure 5. (a) Vertical wind speed and shear vectors in the D4 simulation at 3 km altitude at 10:11 UTC on 30 December 2020. (b) Vertical cross-section YY' showing potential temperature [K] and vertical wind speed. (c) The trajectory of the flight in Figure 3(c). Colors show the recorded vertical acceleration. The small plot shows the acceleration and observed horizontal wind speed during the period of interest.

3.2. Validation of Simulated CAT by Means of a Flight Simulation

We performed 151 virtual flights of a B787 model with a speed of 133 m s^{-1} at an altitude of 3 km with a slightly changing latitude of the initial point in area Z' in Figure 4. Response of the virtual aircraft to turbulence is computed for 100 s for each wind field in each domain with the time fixed at 10:00 UTC.

Figure 6a shows examples of vertical acceleration of the virtual aircraft for each domain. The acceleration changes correspond to the aircraft response to turbulence in simulations D1–D4 during 100 s. Small eddies from the KH wave breaking in simulations D3 and D4 cause indeed strong and rapid shakes for virtual aircraft. Time series of the vertical accelerations recorded by three airplanes are shown in Figure 6c for comparison. The finer is the resolution of the atmospheric simulation, the closer the simulated response is to the observed one. The eddies with a smaller scale effectively shake the aircraft because their fluctuations have frequencies closer to the estimated natural frequency in the pitching motion of the B787 (0.14 Hz, see Text S2 in Supporting Information S1).

The frequency spectra of the simulated (average of 151 spectra)/observed acceleration histories are shown in Figures 6b and 6d. The simulation with finer resolution produces larger amplitudes of shaking at higher

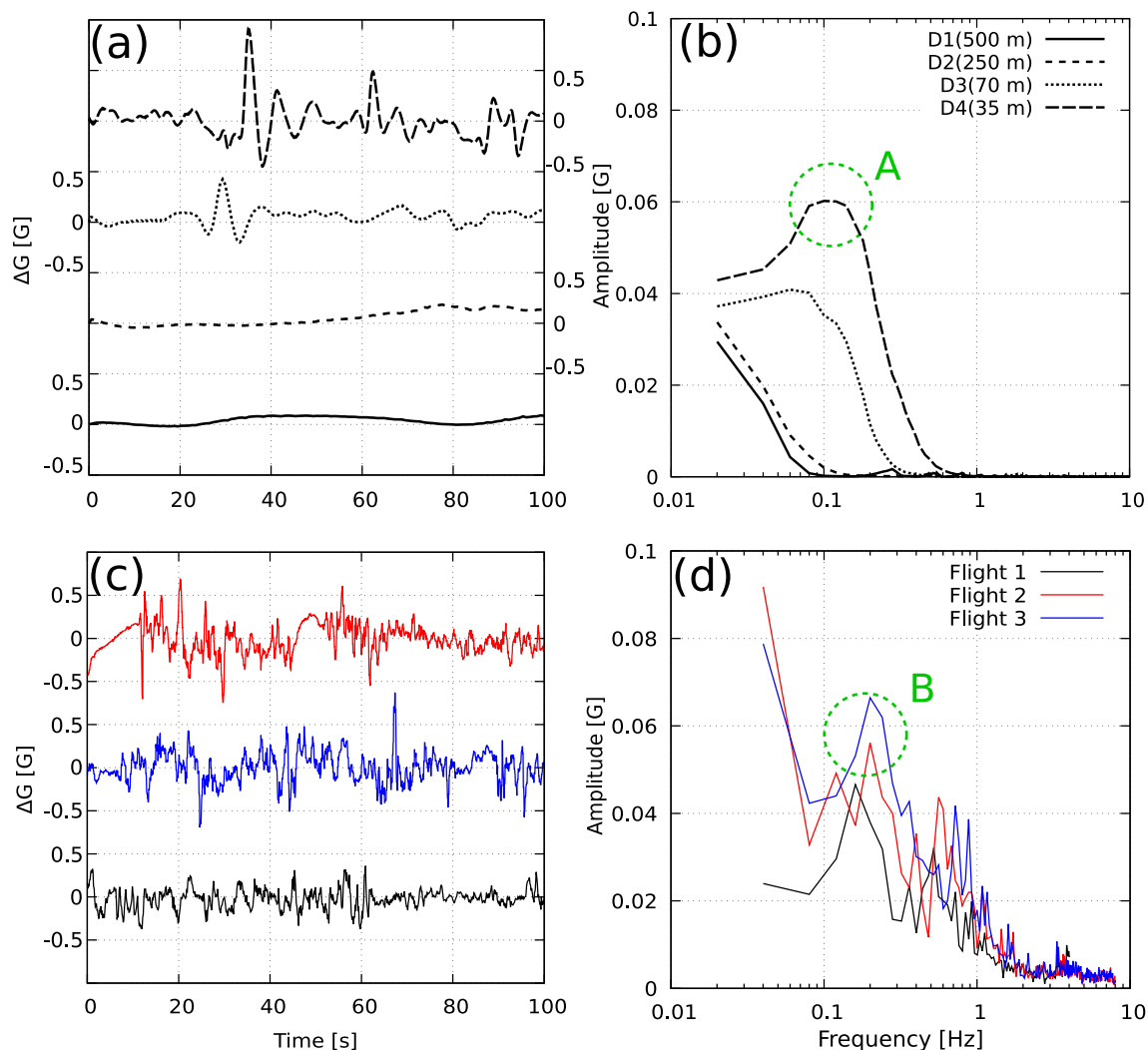


Figure 6. (a) The vertical acceleration histories simulated in D1–D4 and (b) their frequency spectra. Panels (c, d) are the same as (a, b) but for the recorded three acceleration histories.

frequency. The spectrum of virtual flight in simulation D4 is closest to those in the real flight records and has certain amplitudes over most of the range 0.01–0.5 Hz, and peak A is close to the estimated natural frequency in the vertical motion of B787 (0.14 Hz). The observed spectra also have a similar peak (peak B). However, near 1 Hz, the amplitudes decay faster than the observed spectrum. Near-1 Hz oscillation is visible in the flight records (Figure 6c). This difference could be due to the 35 m spacing being near the scale, no pitching rate control in the simulations, and the simulation assumption of a rigid body aircraft (explained in Text S3 in Supporting Information S1). Simulation D4 reasonably reproduced turbulences in the CAT except for those of smaller scale eddies for which interactions with the aircraft itself may not be negligible.

4. Conclusions

CAT occurred over Tokyo and affected many aircrafts on 30 December 2020. An NWP model developed for operational forecasts was employed, and successive downscaling was performed to reproduce the CAT. The strong wind shear satisfied the necessary condition for KH instability, and agreed well with the region where many PIREPs were reported and large shaking was recorded in the flight data. The simulation with 35-m resolution could resolve turbulent eddies with fluctuations from KH wave breaking. The amplitudes of the fluctuations were sufficient to produce “moderate” shaking on a B787-class aircraft without pitching rate control. Comparison

with the time history and spectra of the flight data show that the real/virtual aircrafts were shaken with comparable amplitudes up to one of the natural frequencies of the aircraft motion.

This study verified that the reproduced turbulence was not a mere numerical artifacts but realistic; the signatures of similar turbulence were indeed found in the flight data. Realistic LES in the free atmosphere would be useful for many purposes, which include a better understanding of the turbulence itself and applications in aviation (Sharman & Lane, 2016). Since significant turbulence has been shown to occur in the limited 3D region, routing without flying in the region is possible by adjusting flight levels if the presence of active turbulence is known in advance. LES would provide a smart way of flying if real-time prediction is possible.

Data Availability Statement

GrADS data, flight simulation codes, and outputs are available at https://figshare.com/articles/dataset/data_tar/21152203. ASUCA is available under the collaborative framework between Japan Meteorological Agency and related institutes or universities. However, three flight records and JAL PIREP cannot be made openly available because of confidentiality agreements.

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